

CBE 30355 AI Project 3: Spring '21 Mid Term Solutions

Problem 1: Flow resistance in membranes: unidirectional flows

1. Restatement of the problem

A microfiltration membrane of thickness h is modeled as an array of straight cylindrical pores of radius a . The membrane porosity (fraction open area) is ϵ . For a pressure drop Δp across the membrane thickness h a flow rate per unit membrane area (including both pore and solid area) of Q/A is produced.

- (a) For this model determine the relationship between Δp and Q/A .
- (b) For $a = 2 \mu\text{m}$, $h = 1 \text{ mm}$, $\epsilon = 0.5$, and properties of water, compute the pressure drop required for a flow rate of

$$\frac{Q}{A} = 60 \text{ ml}/(\text{min} \cdot \text{cm}^2).$$

Express the result in atmospheres. (Hint: $1 \text{ dyne}/\text{cm}^2 = 0.99 \times 10^{-6} \text{ atm}$; $1 \mu\text{m} = 10^{-4} \text{ cm}$.)

- (c) Comment quantitatively on whether the assumption of unidirectional, laminar flow is appropriate for cylindrical pores of this size at this flow rate.
- (d) If a thin surface layer has pore radii $a_s = 0.25 \mu\text{m}$, how thin must that layer be so that, at the same Q/A , its pressure drop is equal to the pressure drop of the support layer in part (b)? Give your answer in microns.

2. Detailed step-by-step solution

2.1 Governing equations and assumptions

We assume

- steady, incompressible Newtonian fluid,
- flow is axisymmetric, fully developed, and unidirectional inside each pore: $u_r = u_\theta = 0$, $u_z = u_z(r)$,
- no-slip at the pore wall: $u_z(a) = 0$, and finite velocity on axis: $u_z(0)$ finite,
- pressure varies only in the axial direction inside the pore: $p = p(z)$ and the axial gradient dp/dz is constant along the pore.

Under these assumptions the axial component of the Navier–Stokes equation in cylindrical coordinates reduces to the ordinary differential equation

$$0 = -\frac{dp}{dz} + \mu \left(\frac{1}{r} \frac{d}{dr} \left(r \frac{du_z}{dr} \right) \right),$$

where μ is the dynamic viscosity.

2.2 Integration to obtain the velocity profile (Poiseuille)

Integrate once with respect to r :

$$\frac{d}{dr} \left(r \frac{du_z}{dr} \right) = \frac{1}{\mu} \frac{dp}{dz} r.$$

Integrate again:

$$r \frac{du_z}{dr} = \frac{1}{2\mu} \frac{dp}{dz} r^2 + C_1,$$

so

$$\frac{du_z}{dr} = \frac{1}{2\mu} \frac{dp}{dz} r + \frac{C_1}{r}.$$

Regularity at $r = 0$ requires $C_1 = 0$. Integrate once more:

$$u_z(r) = \frac{1}{4\mu} \frac{dp}{dz} r^2 + C_2.$$

Apply no-slip at $r = a$: $u_z(a) = 0$ gives

$$0 = \frac{1}{4\mu} \frac{dp}{dz} a^2 + C_2,$$

so

$$C_2 = -\frac{1}{4\mu} \frac{dp}{dz} a^2.$$

Thus the familiar parabolic profile (Hagen–Poiseuille) is

$$u_z(r) = -\frac{1}{4\mu} \frac{dp}{dz} (a^2 - r^2).$$

The sign is such that if pressure decreases in the positive z direction ($dp/dz < 0$) then $u_z > 0$.

2.3 Volumetric flow per pore

Volumetric flow through one pore is

$$Q_{\text{pore}} = \int_0^a 2\pi r u_z(r) dr = 2\pi \int_0^a r \left(-\frac{1}{4\mu} \frac{dp}{dz} (a^2 - r^2) \right) dr.$$

Carrying out the integral yields the standard result

$$Q_{\text{pore}} = -\frac{\pi a^4}{8\mu} \frac{dp}{dz}.$$

If the total pressure drop across pore length h is $\Delta p = p_{\text{up}} - p_{\text{down}}$, and assuming a linear drop so $dp/dz = -\Delta p/h$, then

$$Q_{\text{pore}} = \frac{\pi a^4}{8\mu} \frac{\Delta p}{h}.$$

2.4 Relate pore flow to membrane flux Q/A

Let n_p be the number of pores per unit membrane area. The porosity ϵ equals the open area fraction:

$$\epsilon = n_p \times (\text{area of one pore}) = n_p (\pi a^2).$$

Hence

$$n_p = \frac{\epsilon}{\pi a^2}.$$

The volumetric flux per unit membrane area (the given Q/A) is the sum of flows through all pores per unit area:

$$\frac{Q}{A} = n_p Q_{\text{pore}} = \frac{\epsilon}{\pi a^2} \cdot \frac{\pi a^4}{8\mu} \frac{\Delta p}{h} = \frac{\epsilon a^2}{8\mu} \frac{\Delta p}{h}.$$

Rearranging gives the requested relation:

$$\boxed{\Delta p = \frac{8\mu h}{\epsilon a^2} \left(\frac{Q}{A} \right)}.$$

This is the exact algebraic relation for the pore-array membrane model under the stated assumptions.

Equivalently, this can be written in Darcy-like form

$$\frac{\Delta p}{h} = \frac{\mu}{K} \left(\frac{Q}{A} \right) \quad \text{with} \quad K = \frac{\epsilon a^2}{8}$$

so the intrinsic permeability of the pore array is $K = \epsilon a^2/8$.

2.5 Part (b): numerical evaluation

Given:

$$a = 2 \mu\text{m} = 2 \times 10^{-4} \text{ cm}, \quad h = 1 \text{ mm} = 0.1 \text{ cm}, \quad \epsilon = 0.5,$$

and

$$\frac{Q}{A} = 60 \frac{\text{ml}}{\text{min} \cdot \text{cm}^2} = \frac{60 \text{ cm}^3}{60 \text{ s} \cdot \text{cm}^2} = 1.0 \frac{\text{cm}^3}{\text{s} \cdot \text{cm}^2} = 1.0 \text{ cm/s}.$$

Use water viscosity (cgs): $\mu \approx 0.01 \text{ g}/(\text{cm} \cdot \text{s}) = 0.01 \text{ dyn} \cdot \text{s}/\text{cm}^2$.

Plug into

$$\Delta p = \frac{8\mu h}{\epsilon a^2} \left(\frac{Q}{A} \right).$$

Compute numerator and denominator:

$$\Delta p = \frac{8 \times 0.01 \times 0.1}{0.5 \times (2 \times 10^{-4})^2} \times 1.0 = \frac{0.008}{0.5 \times 4 \times 10^{-8}} = \frac{0.008}{2 \times 10^{-8}} = 4.0 \times 10^5 \text{ dyn/cm}^2.$$

Convert to atmospheres using the given conversion $1 \text{ dyn/cm}^2 = 0.99 \times 10^{-6} \text{ atm}$:

$$\Delta p = 4.0 \times 10^5 \times 0.99 \times 10^{-6} \text{ atm} \approx 0.396 \text{ atm}.$$

$$\boxed{\Delta p \approx 4.0 \times 10^5 \text{ dyn/cm}^2 \approx 0.40 \text{ atm}.}$$

2.6 Part (c): is the unidirectional laminar assumption appropriate?

We check the pore-scale Reynolds number. The average axial velocity inside a pore is

$$\bar{u}_{\text{pore}} = \frac{Q_{\text{pore}}}{\pi a^2}.$$

From the relation between membrane flux and pore-average velocity we get the simple identity

$$\bar{u}_{\text{pore}} = \frac{(Q/A)}{\epsilon}.$$

Numerically,

$$\bar{u}_{\text{pore}} = \frac{1.0 \text{ cm/s}}{0.5} = 2.0 \text{ cm/s}.$$

The pore diameter is $D = 2a = 4 \times 10^{-4} \text{ cm}$. Take water density $\rho \approx 1 \text{ g/cm}^3$ and $\mu = 0.01 \text{ g/(cm} \cdot \text{s)}$. Then the Reynolds number based on diameter is

$$\text{Re} = \frac{\rho \bar{u}_{\text{pore}} D}{\mu} = \frac{1 \cdot 2.0 \cdot 4 \times 10^{-4}}{0.01} = 0.08.$$

This is *far* below the usual laminar–turbulent transition ($\text{Re} \sim 2000$), so the flow in each pore is very much in the Stokes (viscous-dominated) regime. Inertia is negligible and the Hagen–Poiseuille solution is appropriate.

Entrance length estimate: for laminar pipe flow a rule-of-thumb for the hydrodynamic entrance length is

$$L_e \approx 0.06 \text{ Re } D.$$

With $\text{Re} = 0.08$ and $D = 4 \times 10^{-4} \text{ cm}$,

$$L_e \approx 0.06 \times 0.08 \times 4 \times 10^{-4} \text{ cm} \approx 1.9 \times 10^{-6} \text{ cm} \approx 1.9 \times 10^{-2} \mu\text{m},$$

which is utterly negligible compared to the pore length $h = 1000 \mu\text{m}$. Thus the fully-developed assumption is safe.

Continuum assumption: the pore radius $2 \mu\text{m}$ is enormous compared to molecular mean free paths in liquids, so continuum hydrodynamics applies.

Conclusion: the unidirectional, laminar, fully-developed assumption is quantitatively justified for the numbers in (b).

2.7 Part (d): thin surface layer with $a_s = 0.25 \mu\text{m}$

Using the derived formula

$$\Delta p = \frac{8\mu h}{\epsilon a^2} \left(\frac{Q}{A} \right),$$

and assuming the surface layer has the *same* porosity ϵ and is at the same flux Q/A , the pressure drop across the surface layer of thickness h_s is proportional to h_s/a_s^2 . To make the surface-layer pressure drop equal to the support-layer pressure drop (the layer from part (b)), set

$$\frac{h_s}{a_s^2} = \frac{h_{\text{support}}}{a_{\text{support}}^2}.$$

Hence

$$h_s = h_{\text{support}} \left(\frac{a_s}{a_{\text{support}}} \right)^2.$$

With $h_{\text{support}} = 1 \text{ mm} = 1000 \text{ }\mu\text{m}$, $a_{\text{support}} = 2 \text{ }\mu\text{m}$ and $a_s = 0.25 \text{ }\mu\text{m}$:

$$h_s = 1000 \text{ }\mu\text{m} \times \left(\frac{0.25}{2}\right)^2 = 1000 \text{ }\mu\text{m} \times (0.125)^2 = 1000 \text{ }\mu\text{m} \times 0.015625 = 15.625 \text{ }\mu\text{m}.$$

$h_s \approx 15.6 \text{ }\mu\text{m}.$

So the surface layer need only be on the order of tens of microns thick to contribute a pressure drop comparable to the 1 mm support layer when its pores are eight times smaller in radius.

3. Study guide related to this problem

3.a Key concepts examined

1. **Derivation of Poiseuille flow from the Navier–Stokes equations:** converting the axial momentum equation in cylindrical coordinates to the simple ODE and integrating to obtain the parabolic velocity profile and $Q \propto a^4$ law.
2. **Connection between pore-scale flow and macroscopic flux:** using porosity to convert single-pore Hagen–Poiseuille flow into a membrane flux Q/A , and the resulting expression for permeability $K = \epsilon a^2/8$ (Darcy-like form).
3. **Dimensionless numbers and scaling checks:** estimating Reynolds number, entrance length, and checking continuum assumptions to justify model assumptions.

3.b References for further reading (book, class notes, verified web source)

- **Textbook (related problems):** R. B. Bird, W. E. Stewart, and E. N. Lightfoot, *Transport Phenomena*, 2nd ed., Wiley. See the sections on viscous flow / Poiseuille problems (examples and problem set: §2–3 and the problem set the book lists as 2B.x; a common related exercise is Problem 2B.3, the slit analog of Hagen–Poiseuille). (Book: Bird, Stewart & Lightfoot.)
- **Course lecture notes (class notes):** The course notes for CBE 30355 (Transport I) include a lecture “Poiseuille Flow, Couette Flow” (Sept. 30, 2025). Online notes: <https://diffusivepress.org/courseware/cbe30355/notes/111/notes.html>
- **Verified web source (derivation and discussion):**
 - MIT OCW / lecture notes: Couette & Poiseuille flows (derivation from NS). Example: <https://ocw.mit.edu/courses/2-25-advanced-fluid-mechanics-fall-2013/>
 - NPTEL lecture on Hagen–Poiseuille flow (derivation and assumptions): https://archive.nptel.ac.in/content/storage2/courses/112104118/lecture-26/26-3_hag_poiseuille.htm
 - Wikipedia (Hagen–Poiseuille equation) for a quick formula summary: https://en.wikipedia.org/wiki/Hagen%E2%80%93Poiseuille_equation

3.c Suggested related practice problem

Practice: Consider a membrane whose pores are circular but with a distribution of radii. Suppose half the open area consists of pores of radius a_1 and half of the open area consists of pores of radius a_2 (same total porosity ϵ). Derive an expression for the overall membrane flux Q/A as a function of Δp , h , a_1 , a_2 , and ϵ . Compute the fractional contribution to flux from the small pores when $a_2 = 2a_1$.

4. Quick summary of main formulae (for reference)

$$u_z(r) = -\frac{1}{4\mu} \frac{dp}{dz} (a^2 - r^2), \quad Q_{\text{pore}} = \frac{\pi a^4}{8\mu} \frac{\Delta p}{h},$$

$$\Delta p = \frac{8\mu h}{\epsilon a^2} \left(\frac{Q}{A} \right), \quad K = \frac{\epsilon a^2}{8}.$$

Notes on units: the algebra above used cgs units (cm, s, dyn) because the hint and intermediate manipulations are convenient in cgs. Converting to SI is straightforward via the usual factors.

5. Video Narration

The narration of the solution to this problem is found at the url:

<https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=ca17c180-70cf-45c4-a6f4-b36e>

Problem 2: Dimensional Analysis and Scaling

1. Restatement of the problem

We examine mixing in the H. B. Fuller PB01 mixing tank under *strict dynamic similarity*. The prototype (full scale) tank has radius $R_{\text{full}} = 1.00$ m and height $H_{\text{full}} = 3.00$ m. The impeller rotates at 40 rpm. The fluid being mixed has dynamic viscosity $\mu_{\text{full}} \approx 200$ poise and density $\rho_{\text{full}} = 1.00$ g/cm³. A 1:9 scale model (linear scale factor $s = L_{\text{model}}/L_{\text{full}} = 1/9$) is proposed. Required:

- (a) What should be the properties of the simulant in the scale model?
- (b) What should be the rotation speed of the impeller in the scale model?
- (c) If an encoder measures torque on the model shaft, how does that torque scale to the full-scale torque?
- (d) For interfacial transport (free-surface present), if the diffusion coefficient of water in the full-scale viscous fluid is $D_{\text{full,w}} = 1 \times 10^{-6}$ cm²/s, what diffusion coefficient must the water-simulant have in the model to preserve strict dynamic similarity?

2. Solution

Assumptions and notation

- Geometric similarity is exact; the model is a geometric scale $s = 1/9$ of the prototype.
- We assume a typical industrial impeller diameter D equal to one third of the tank diameter (a common design rule): thus

$$D_{\text{full}} = \frac{1}{3} \times (2R_{\text{full}}) = \frac{2}{3} \text{ m} \approx 0.6666 \text{ m}.$$

(This is an explicit assumption; if you have a different D_{full} replace the numerical value below and the algebra still applies.)

- Use these symbols:

$$s = \frac{L_{\text{model}}}{L_{\text{full}}} = \frac{1}{9},$$

N = impeller rotation rate (rev/s),

ρ , μ , ν = density, dynamic viscosity, kinematic viscosity ($\nu = \mu/\rho$),

D = impeller diameter (length scale for the impeller).

- Gravity $g = 9.81$ m/s². Units are SI unless otherwise stated; when convenient cgs units are shown (poise, cm, etc.) because the problem statement used poise and g/cm³.

Key dimensionless numbers to preserve (because free surface is present)

For strict dynamic similarity with a free surface and with mass transport, the relevant dimensionless groups that must be matched are:

- **Reynolds number (Re)** – ratio of inertial to viscous forces (here using impeller-based scale)

$$Re = \frac{\rho N D^2}{\mu}.$$

- **Froude number (Fr)** – ratio of inertial (centrifugal) acceleration to gravity relevant to free-surface effects

$$Fr = \frac{N^2 D}{g}.$$

- **Peclet number (Pe)** for mass transport (water transport by advection versus diffusion)

$$Pe = \frac{UL}{D_{\text{diff}}} \sim \frac{(ND)D}{D_{\text{diff}}} = \frac{ND^2}{D_{\text{diff}}}.$$

To enforce strict dynamic similarity we will require

$$Re_{\text{model}} = Re_{\text{full}}, \quad Fr_{\text{model}} = Fr_{\text{full}}, \quad Pe_{\text{model}} = Pe_{\text{full}}.$$

Algebraic derivation of scaling laws

Start from geometric scaling:

$$D_{\text{model}} = s D_{\text{full}}.$$

Preserve Froude number:

$$N_{\text{model}}^2 D_{\text{model}} = N_{\text{full}}^2 D_{\text{full}} \Rightarrow N_{\text{model}} = \frac{N_{\text{full}}}{\sqrt{s}}. \quad (1)$$

Preserve Reynolds number:

$$\frac{\rho_{\text{model}} N_{\text{model}} D_{\text{model}}^2}{\mu_{\text{model}}} = \frac{\rho_{\text{full}} N_{\text{full}} D_{\text{full}}^2}{\mu_{\text{full}}}.$$

Substitute $D_{\text{model}} = s D_{\text{full}}$ and $N_{\text{model}} = N_{\text{full}}/\sqrt{s}$ and cancel common factors to obtain

$$\frac{\rho_{\text{model}}}{\mu_{\text{model}}} s^{3/2} = \frac{\rho_{\text{full}}}{\mu_{\text{full}}}.$$

Rearrange using kinematic viscosity $\nu = \mu/\rho$:

$$\nu_{\text{model}} = \nu_{\text{full}} s^{3/2}. \quad (2)$$

Preserve Peclet number (for diffusion):

$$\frac{N_{\text{model}} D_{\text{model}}^2}{D_{\text{model,diff}}} = \frac{N_{\text{full}} D_{\text{full}}^2}{D_{\text{full,diff}}}.$$

Using the same substitutions as above leads to

$$D_{\text{model,diff}} = D_{\text{full,diff}} s^{3/2}. \quad (3)$$

Finally, torque scaling. Dimensional analysis for torque T (units $M L^2/T^2$) for an agitated tank gives the common inertial form

$$T \sim \rho N^2 D^5,$$

while the viscous (creeping flow) form is

$$T \sim \mu N D^3.$$

If Re and Fr are preserved and geometry is similar, both forms yield the same *ratio* of model to prototype torque. Using either form and substituting the scaling relations gives

$$\frac{T_{\text{model}}}{T_{\text{full}}} = \frac{\rho_{\text{model}}}{\rho_{\text{full}}} s^4. \quad (4)$$

Thus for equal densities the torque scales as s^4 (i.e. $T_{\text{model}} = s^4 T_{\text{full}}$), or equivalently

$$T_{\text{full}} = \frac{T_{\text{model}}}{s^4}.$$

Numerical evaluation (with our assumption $D_{\text{full}} = \frac{2}{3}$ m)

First compute a few prototype quantities.

Prototype impeller quantities

$$D_{\text{full}} = \frac{2}{3} \text{ m} \approx 0.6666667 \text{ m},$$

$$N_{\text{full}} = 40 \text{ rpm} = \frac{40}{60} \text{ rev/s} = \frac{2}{3} \text{ rev/s} \approx 0.6666667 \text{ s}^{-1}.$$

Fluid properties given:

$$\mu_{\text{full}} = 200 \text{ poise} = 200 \times 0.1 \text{ Pa} \cdot \text{s} = 20 \text{ Pa} \cdot \text{s},$$

$$\rho_{\text{full}} = 1.00 \text{ g/cm}^3 = 1000 \text{ kg/m}^3.$$

Hence

$$\nu_{\text{full}} = \frac{\mu_{\text{full}}}{\rho_{\text{full}}} = \frac{20}{1000} = 0.02 \text{ m}^2/\text{s} = 200 \text{ cm}^2/\text{s}.$$

Compute prototype Reynolds number (impeller-based)

$$Re_{\text{full}} = \frac{\rho_{\text{full}} N_{\text{full}} D_{\text{full}}^2}{\mu_{\text{full}}} \approx \frac{1000 \times 0.6666667 \times (0.6666667)^2}{20} \approx 14.815.$$

So the prototype is in a low- Re (viscous-dominated / laminar) regime. Compute prototype Froude number

$$Fr_{\text{full}} = \frac{N_{\text{full}}^2 D_{\text{full}}}{g} \approx \frac{(0.6666667)^2 \times 0.6666667}{9.81} \approx 0.0302.$$

Model geometry and rotation rate

$$s = \frac{1}{9}, \quad D_{\text{model}} = s D_{\text{full}} = \frac{1}{9} \times 0.6666667 \text{ m} \approx 0.0740741 \text{ m} (7.4074 \text{ cm}).$$

From (1):

$$N_{\text{model}} = \frac{N_{\text{full}}}{\sqrt{s}} = N_{\text{full}} \times 3 = 3 \times 0.6666667 \text{ s}^{-1} = 2.0 \text{ rev/s} = 120 \text{ rpm}.$$

So the model impeller must run at **120 rpm**.

Model kinematic and dynamic viscosity (part a) From (2):

$$\nu_{\text{model}} = \nu_{\text{full}} s^{3/2}.$$

Since $s^{3/2} = (1/9)^{3/2} = 1/27 \approx 0.037037037$,

$$\nu_{\text{model}} = 200 \text{ cm}^2/\text{s} \times \frac{1}{27} \approx 7.407407 \text{ cm}^2/\text{s}.$$

With $\rho_{\text{model}} = \rho_{\text{full}} = 1.00 \text{ g/cm}^3$ this gives

$$\mu_{\text{model}} = \rho_{\text{model}} \nu_{\text{model}} \approx 7.4074 \text{ poise} \approx 0.7407407 \text{ Pa} \cdot \text{s}.$$

Answer to (a): The model simulant should have (approximately)

$$\rho_{\text{model}} = 1.00 \text{ g/cm}^3, \quad \mu_{\text{model}} \approx 7.41 \text{ poise (0.741 Pa} \cdot \text{s)},$$

i.e. a kinematic viscosity $\nu_{\text{model}} \approx 7.41 \text{ cm}^2/\text{s}$. (If you prefer a different model density you can keep ν_{model} fixed and adjust $\mu_{\text{model}} = \rho_{\text{model}} \nu_{\text{model}}$.)

Answer to (b) Rotation speed in the model:

$$N_{\text{model}} = 120 \text{ rpm}.$$

Answer to (c) Torque scaling From (4) with $\rho_{\text{model}} = \rho_{\text{full}}$:

$$\frac{T_{\text{model}}}{T_{\text{full}}} = s^4 = \left(\frac{1}{9}\right)^4 = \frac{1}{6561}.$$

Hence

$$T_{\text{full}} = 6561 T_{\text{model}}.$$

So the full-scale torque is larger than the model torque by a factor of 6561. Conversely, $T_{\text{model}} = T_{\text{full}}/6561$.

Answer to (d) Diffusion coefficient for the model From (3):

$$D_{\text{model,diff}} = D_{\text{full,diff}} s^{3/2} = D_{\text{full,diff}}/27.$$

Given $D_{\text{full,diff}} = 10^{-6} \text{ cm}^2/\text{s}$,

$$D_{\text{model,diff}} = \frac{10^{-6}}{27} \text{ cm}^2/\text{s} \approx 3.7037 \times 10^{-8} \text{ cm}^2/\text{s}.$$

Practical remarks and feasibility

- The required model dynamic viscosity $\mu_{\text{model}} \approx 7.4 \text{ poise}$ is large compared with water (0.01 poise) but much smaller than the prototype 200 poise. In practice an aqueous glycerol (glycerol) solution or a syrupy polymer solution can provide a tuneable viscosity near this value. (See the glycerol–water viscosity tables in the references.)
- The diffusion coefficient required in (d) is extremely small (about $3.7 \times 10^{-8} \text{ cm}^2/\text{s}$). Molecular diffusion coefficients in liquids are typically 10^{-5} – $10^{-7} \text{ cm}^2/\text{s}$ for small molecules; 3.7×10^{-8} is at the low end (requires either very large molecules, gels, or effective “apparent” diffusivity slowed by microstructure). Consequently, matching molecular diffusion exactly in a small lab model may be impractical. Two practical approaches:

1. Use a *surrogate tracer* whose effective diffusivity in the model fluid matches the required $D_{\text{model,diff}}$ (e.g. large dye molecules, colloids, or polymer tracers), or
 2. Accept a mismatch in molecular D and use dimensionless correlations (Sherwood vs Re , Sc) or scale-up models to predict prototype mass transfer from measured model data; in doing so you must account for the Schmidt number $Sc = \nu/D_{\text{diff}}$ and the resulting effect on Sherwood number.
- Because the prototype Reynolds number is low ($Re \approx 15$), the flow is viscous-dominated; this reduces some complications (no fully developed turbulence), but it also means torque and mixing rates are sensitive to exact geometry and clearance effects (endwalls, shaft seals, small geometric imperfections) in the model. Careful fabrication is required.

3. Study guide

1. Key concepts examined by this problem

- Dynamic similarity and geometric similarity: how to use dimensionless groups to design scale models (Re , Fr , Pe).
- Scaling of transport coefficients (viscosity, diffusivity) and operational parameters (rotation rate) from prototype to model.
- Torque and mechanical-power scaling for rotating equipment and practical limitations when diffusion is molecular-limited.

2. References for further reading (book, class notes, web)

- Bird, R. B., Stewart, W. E., & Lightfoot, E. N., *Transport Phenomena*, revised edition. See Section 3.7 “Dimensional Analysis of the Equations of Change” and the exercises at the end of the chapter for worked exercises on similarity and scaling (use chapter 3/3.7 in the edition you have).
- Class notes (CBE 30355, Prof. D. T. Leighton) — dynamic similarity lecture (Oct. 7) and dimensional analysis lecture (Oct. 9).
<https://diffusivepress.org/courseware/cbe30355/notes/113/notes.html>
- MIT OpenCourseWare on dimensional analysis and the Buckingham π theorem:
<https://ocw.mit.edu/courses/2-25-advanced-fluid-mechanics-fall-2013/pages/dimensional-analysis/>
- Practical viscosity tables for aqueous glycerol (useful to pick a simulant with $\mu \approx 7.4$ poise).

3. Related practice problem (suggestion)

Design a 1:4 scale model for the same tank (same geometric assumption $D = 1/3$ of tank diameter). For a prototype fluid with $\mu_{\text{full}} = 50$ poise and $\rho_{\text{full}} = 1.0 \text{ g/cm}^3$ and prototype rotation 30 rpm, compute the required model fluid viscosity, model rotation speed, torque scaling factor, and the required model diffusion coefficient if $D_{\text{full,diff}} = 5 \times 10^{-7} \text{ cm}^2/\text{s}$. Compare viscous- and inertial-dominated torque predictions and comment on feasibility.

4. Video Narration

The narration of the solution to this problem is found at the url:

<https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=8692ddf9-4363-400e-b627-b36e>

Problem 3: Hydrostatics / Archimedes' Law

1. Restatement of the problem

A hydrometer is made by attaching an air-filled straw (negligible mass) to an acrylic sphere. The sphere has density $\rho_s = 1.18 \text{ g/cm}^3$ and radius $R = 1.50 \text{ cm}$. The straw has radius $r = 0.25 \text{ cm}$ and unknown length L .

- (a) What length L of straw is *necessary* so that the device can be used to measure the density of (fresh) water with $\rho_f = 1.00 \text{ g/cm}^3$? (Interpretation: find the minimum straw length so that the assembly can float in water; equivalently the straw length required so that, if the straw is fully submerged together with the sphere, the buoyant volume is sufficient to balance the weight of the acrylic sphere.)
- (b) If the actual fluid has density $\rho_{sw} = 1.024 \text{ g/cm}^3$ (typical saltwater salinity target), what length of the straw will stick out of the water for the straw length found in part (a)?

2. Solution (step-by-step)

We apply Archimedes' principle: at equilibrium the buoyant force equals the weight of the hydrometer. Because the straw is air-filled and negligible mass, the weight is that of the acrylic sphere alone. The buoyant force equals the weight of the displaced fluid, i.e. $\rho_f g$ times the submerged volume.

Geometry and symbols

$$\begin{aligned} R &= 1.50 \text{ cm} \quad (\text{sphere radius}), \\ \rho_s &= 1.18 \text{ g/cm}^3 \quad (\text{sphere material density}), \\ r &= 0.25 \text{ cm} \quad (\text{straw radius}), \\ A &= \pi r^2 \quad (\text{straw cross-sectional area}), \\ V_s &= \frac{4}{3}\pi R^3 \quad (\text{sphere volume}), \\ m &= \rho_s V_s \quad (\text{mass of sphere}), \\ \rho_f & \quad (\text{fluid density; will be } 1.000 \text{ or } 1.024 \text{ g/cm}^3). \end{aligned}$$

(a) Minimum straw length to float in fresh water

If the straw is long enough that the maximum displaced volume available (sphere plus entire straw volume) must be at least the displaced volume required to support the sphere, then the minimal length L satisfies

$$\rho_f (V_s + AL) = m = \rho_s V_s.$$

Solve for L :

$$L = \frac{\rho_s V_s / \rho_f - V_s}{A} = \frac{V_s}{A} \left(\frac{\rho_s}{\rho_f} - 1 \right).$$

Now substitute the numeric values (all in cgs units so units are cm and g/cm³):

$$V_s = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(1.50)^3 = \frac{4}{3}\pi(3.375) = 4.5\pi \text{ cm}^3,$$

$$A = \pi r^2 = \pi(0.25)^2 = 0.0625\pi \text{ cm}^2.$$

Note that the factor π cancels in the ratio V_s/A . With $\rho_f = 1.00 \text{ g/cm}^3$:

$$L = \frac{4.5\pi}{0.0625\pi} \left(\frac{1.18}{1.00} - 1 \right) = \frac{4.5}{0.0625} \times 0.18,$$

$$\frac{4.5}{0.0625} = 72.0, \quad 72.0 \times 0.18 = 12.96 \text{ cm}.$$

Answer to (a). The minimum straw length is

$L \approx 12.96 \text{ cm (use 13.0 cm to two significant figures).}$

(b) Straw length sticking out in salt water of density 1.024 g/cm³

For the salt water density $\rho_{sw} = 1.024 \text{ g/cm}^3$ the submerged volume required to support the sphere is

$$V_{\text{sub}} = \frac{m}{\rho_{sw}} = \frac{\rho_s V_s}{\rho_{sw}}.$$

The submerged portion of the straw (call its submerged length s) then satisfies

$$V_{\text{sub}} = V_s + As \implies s = \frac{V_{\text{sub}} - V_s}{A} = \frac{V_s}{A} \left(\frac{\rho_s}{\rho_{sw}} - 1 \right).$$

Evaluate numerically:

$$\frac{\rho_s}{\rho_{sw}} = \frac{1.18}{1.024} = 1.15234375,$$

so

$$s = \frac{4.5}{0.0625} \times (1.15234375 - 1) = 72.0 \times 0.15234375 = 10.96875 \text{ cm}.$$

Therefore the length of straw *sticking out* of the water is

$$\Delta = L - s = 12.96 - 10.96875 = 1.99125 \text{ cm} \approx 1.99 \text{ cm}.$$

Answer to (b). With the straw length from part (a) the portion sticking out in $\rho = 1.024 \text{ g/cm}^3$ water is

$\Delta \approx 1.99 \text{ cm (about 2.0 cm).}$

Quick numeric summary

$$V_s = 4.5\pi \text{ cm}^3 \approx 14.137 \text{ cm}^3,$$

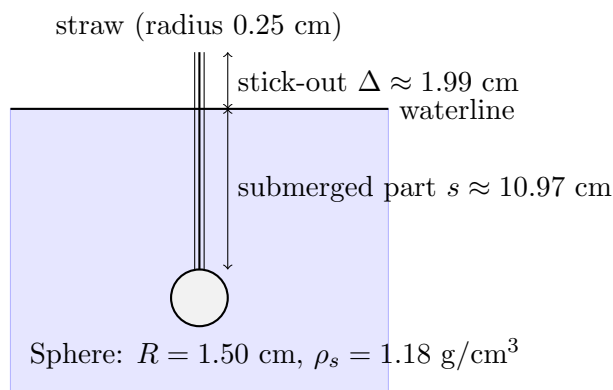
$$m = \rho_s V_s \approx 1.18 \times 14.137 \approx 16.682 \text{ g},$$

$$A = 0.0625\pi \text{ cm}^2 \approx 0.19635 \text{ cm}^2,$$

$$L_{\text{min}} \approx 12.96 \text{ cm},$$

sticking-out in $\rho = 1.024 \approx 1.99 \text{ cm}.$

TikZ schematic (not to scale for readability)



Straw radius $r = 0.25 \text{ cm}$; chosen length $L \approx 12.96 \text{ cm}$

3. Study guide

1. Key concepts examined (2–3 items).

- Archimedes' principle / buoyancy: the buoyant force equals the weight of the displaced fluid.
- Conservation of mass (as applied to densities and displaced volume) and simple geometric volume calculations (sphere, cylinder).
- Dimensional consistency and unit handling (here we used cgs: cm and g).

2. References for further study (book, class notes, and a verified web source).

- Bird, R. B., Stewart, W. E., and Lightfoot, E. N., *Transport Phenomena*, While a classic text, there are no examples relating to Archimedes Law hydrostatics problems. Integral momentum problems are provided in Chapter 7.
- Class notes (lecture on Hydrostatics) for this course: “CBE 30355 – Lecture Notes, Sept. 2, 2025 – Hydrostatics”. See the course page at DiffusivePress for the Hydrostatics lecture (includes worked material and links to the narrated lecture).<https://diffusivepress.org/courseware/cbe30355/notes/103/notes.html>
- Britannica: article “Archimedes’ principle” for historical and concise physical statement. <https://www.britannica.com/science/Archimedes-principle>

3. Suggested related practice problem.

Design a hydrometer for measuring a range of densities from 0.990 to 1.030 g/cm^3 using the same acrylic sphere. Determine the optimal straw length and a simple graduated scale (marks on the straw) that converts the observed stick-out length to fluid density. Show the derivation and provide the scale values at increments of 0.005 in density.

This practice problem extends the algebra used above and gives practice with linear mappings between displaced volume and density, and with practical design choices (scale readability, choice of straw radius).

4. Video Narration

The narration of the solution to this problem is found at the url:

<https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=8454c5ce-8d1f-4eb1-955f-b36e>

Problem 4: Hydroelectric Power at Seitz Park

1. Restatement of the problem

The University installs a hydroelectric generation system on the St. Joseph River at Seitz Park (East Race). The river's average flow rate is 3000 cfs, but 600 cfs is reserved for other purposes and must be subtracted. The available head (elevation drop) is approximately 3 m.

- a. With turbine efficiency $\eta = 0.80$, compute the expected electrical power output in megawatts (MW). (Unit conversion: $1 \text{ ft}^3 = 28.3168466 \text{ L} = 0.0283168466 \text{ m}^3$.)
- b. Using the electricity price 0.12 \$/kWh, compute the annual monetary value of the electricity produced in one year.

2. Solution (step-by-step)

2.1 Relevant formula

The electric power available from falling water (accounting for turbine efficiency) is

$$P = \eta \rho g Q H,$$

where

- P is electrical power (W),
- η is turbine efficiency (dimensionless),
- ρ is density of water (kg/m^3),
- g is acceleration due to gravity (m/s^2),
- Q is volumetric flow rate through the turbine (m^3/s),
- H is head (m).

2.2 Given / assumed values

Average river flow = 3000 cfs,

Reserved flow = 600 cfs,

$\Rightarrow$ Available flow for turbines $Q_{\text{cfs}} = 3000 - 600 = 2400 \text{ cfs}$.

$H = 3.0 \text{ m}$,

$\eta = 0.80$,

$\rho \approx 1000 \text{ kg/m}^3$,

$g \approx 9.81 \text{ m/s}^2$.

2.3 Convert the flow to SI units

Use $1 \text{ cfs} = 0.0283168466 \text{ m}^3/\text{s}$.

$$Q = 2400 \text{ cfs} \times 0.0283168466 \frac{\text{m}^3}{\text{s} \cdot \text{cfs}} = 2400 \times 0.0283168466 \text{ m}^3/\text{s}.$$

Compute:

$$Q = 67.9604318 \text{ m}^3/\text{s} \quad (\text{rounded to 9 significant figures}).$$

2.4 Compute the electric power

Now substitute into $P = \eta \rho g Q H$:

$$\begin{aligned} P &= 0.80 \times (1000 \text{ kg/m}^3) \times (9.81 \text{ m/s}^2) \times (67.9604318 \text{ m}^3/\text{s}) \times (3.0 \text{ m}) \\ &= 1\,600\,060.4 \text{ W} \quad (\text{approximately}). \end{aligned}$$

Convert to megawatts:

$$P \approx 1.60006 \text{ MW}.$$

Rounded sensibly:

$$P \approx 1.60 \text{ MW}$$

(Three significant figures appropriate for given data.)

2.5 Annual energy and monetary value

Electrical energy produced per year (assuming continuous operation at the above power) is

$$E_{\text{year}} = P \times t_{\text{year}} = 1\,600\,060.4 \text{ W} \times (365 \times 24 \text{ h})$$

Convert watts to kilowatts for kWh:

$$E_{\text{year}} = \frac{1\,600\,060.4}{1000} \text{ kW} \times 8760 \text{ h} = 1\,4016\,529.16 \text{ kWh} \quad (\text{approximately } 1.4017 \times 10^7 \text{ kWh}).$$

Using price $0.12 \text{ \$/kWh}$,

$$\text{Revenue per year} = E_{\text{year}} \times 0.12 \frac{\$}{\text{kWh}} \approx 1\,681\,983.50 \$.$$

Rounded to three significant digits:

$$\text{Annual value} \approx 1.68 \times 10^6 \text{ dollars per year}.$$

2.6 Summary of numerical answers

- Expected electrical power output: 1.60 MW .
- Expected monetary value of electricity (at $\$0.12/\text{kWh}$): $\$1.68 \times 10^6 \text{ per year}$.

3. Study guide (requested elements)

a. Key concepts examined by this problem

1. **Energy balance for steady flow:** recognizing that the potential energy flux of water, $\rho g Q H$, is the source of mechanical power available to the turbine and (after efficiency) electrical power.
2. **Unit conversions and dimensional consistency:** accurate conversion from cubic feet per second to cubic meters per second and careful unit-tracking to produce watts and kWh.
3. **Efficiency and real systems:** accounting for turbine/generator efficiency reduces theoretical hydraulic power to delivered electrical power; understanding continuous-operation assumptions when computing annual energy.

b. References for further study

- Bird, R. B., Stewart, W. E., and Lightfoot, E. N., *Transport Phenomena*. The most closely related sections are in Chapter 7, particularly section 7.4.
- Course notes (class web page and index) for CBE 30355 (Transport I) with lecture notes and related material:
<https://diffusivepress.org/courseware/cbe30355>. These notes include discussions of hydrostatics, energy-like arguments, and example problems useful for this sort of flow-power calculation. The most closely related lecture is on conservation of momentum, <https://diffusivepress.org/courseware/cbe30355/notes/107/notes.html>
- Hydropower fundamentals (derivation and examples for $P = \eta \rho g Q H$): U.S. Department of Energy, “Hydropower Basics” and companion pages. Useful for context and typical efficiency ranges.
<https://www.energy.gov/eere/water/hydropower-basics>.

c. Suggested related practice problem

Practice problem: Suppose the East Race project can operate as a run-of-river plant with seasonal variation in flow. If the available flow Q varies between 1200 cfs (low season) and 3000 cfs (high season), with the same 3 m head and $\eta = 0.80$, compute (i) the electrical power at low and high flows, (ii) the energy produced in a year if the plant runs at the low-flow value for 6 months and high-flow value for 6 months, and (iii) the annual revenue at \$0.12/kWh. Discuss how installing a small upstream retention (pond) that increases head to $H = 4$ m during low season would change annual production and revenue.

4. Video Narration

The narration of the solution to this problem is found at the url:

<https://notredame.hosted.panopto.com/Panopto/Pages/Viewer.aspx?id=ca17c180-70cf-45c4-a6f4-b36e>

AI Interaction Notes

1 Main Prompt

The primary prompt used was:

Act as an agent, solving the following problem in a step by step manner to be sure that we have the correct solution.

Return the answer as a LaTeX script in Chat for compilation without errors in TeXShop.

Do not use unicode characters, ascii safe only.

Any necessary figures should be given as a tikz script.

The problem is taken from an old Transport midterm examination, closed books and notes.

Your report on this problem should have the following elements:

- 1) a restatement of the problem.
- 2) a detailed step-by-step solution to the problem.
- 3) A study guide related to this problem.

The study guide should contain the following elements:

- a. 2 or 3 key concepts that are being examined in this problem.
- b. References to related problems in Bird, Stewart & Lightfoot (section and problem number), the url of the appropriate lecture from the class notes (the class notes webpage and index is at <https://diffusivepress.org/courseware/cbe30355>), and a verified web source for further study with url.
- c. A suggestion for a related problem for further practice

The problem is:

[paste problem here]

Problem 1 Issues

For the first problem a line was added:

(note: for this problem derive Poiseuille's Law from the NS equations in cylindrical coordinates)

In addition, the title was modified, and the TikZ diagram (which was better than usual) was removed rather than edited. The URLs were verified. Chat had added a section on references which was redundant and removed, as were its usual "if you like I can" verbiage at the end.

Problem 2 Issues

For the second problem it was pointed out that there was a free surface so that it would preserve Fr. The file generated by the AI was partially interpreted in Chat. When asked to provide the script to the LaTeX a much inferior version was generated - it lost a large fraction of its analysis. The (much better) original version was pasted back into Chat with the instruction to convert it into a latex script. This hung up as well, however only the last few lines were lost (and not needed except for an enddocument line which was manually added in).

Problem 3 Issues

The third problem was done correctly. A latex error was corrected and a few minor changes were made. The urls were provided in a references section, and were a mixed bag. The reference to the class lecture was correct, however the reference to BSL was halucinated as the text doesn't actually cover Archimedes Law and hydrostatics. Modified urls and references were corrected. A tikz plot was provided (although it was not necessary) was kept, and some edits to it were made.

Problem 4 Issues

The fourth problem was calculated correctly. The solution was edited, eliminating the TikZ picture. The AI was unable to identify an appropriate section in BSL or the correct lecture from class, so these were added in manually.

Final Edits

The four problem solutions were combined into a single LaTeX file, the Quicktime recordings were made and uploaded to Panopto. The links were then imbedded into the final project report.